

Event-based Stereo and Motion-related Tasks

Prof. Dr. Guillermo Gallego

guillermo.gallego@tu-berlin.de

<http://www.guillermogallego.es>

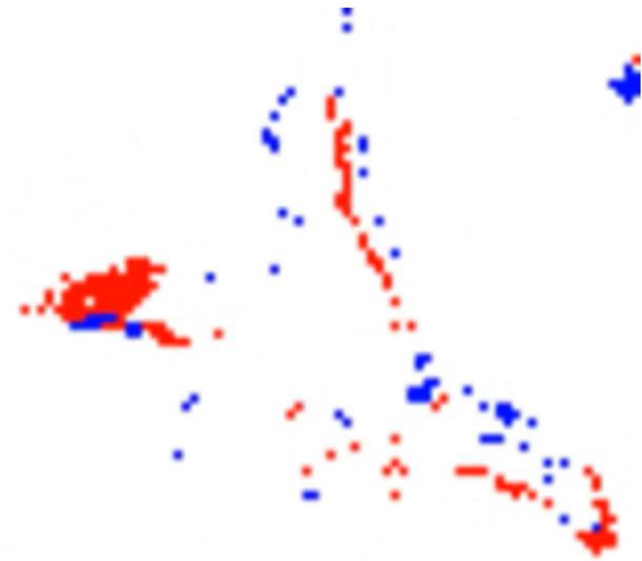
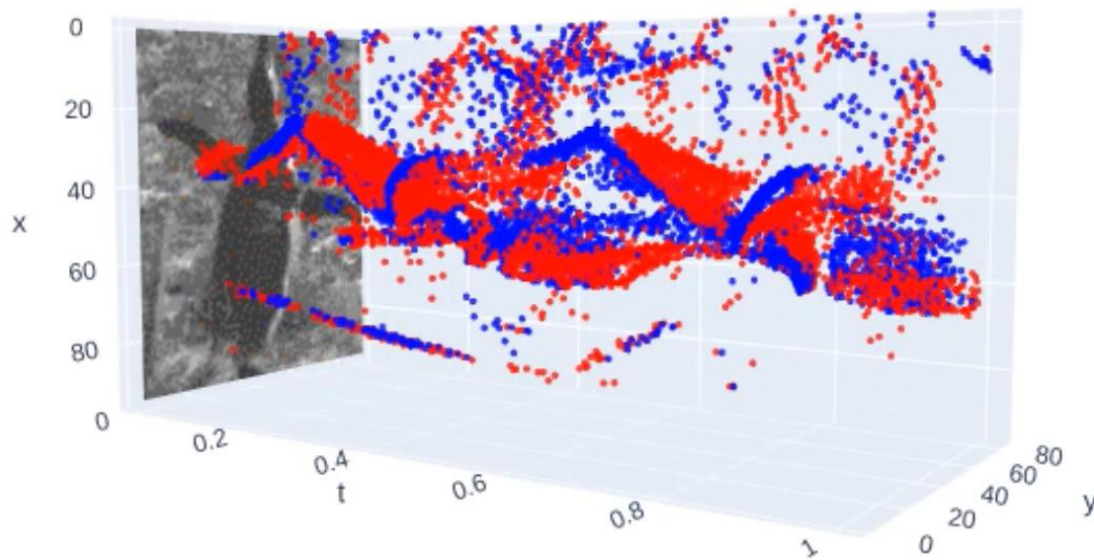
Event cameras for animal monitoring

Dataset

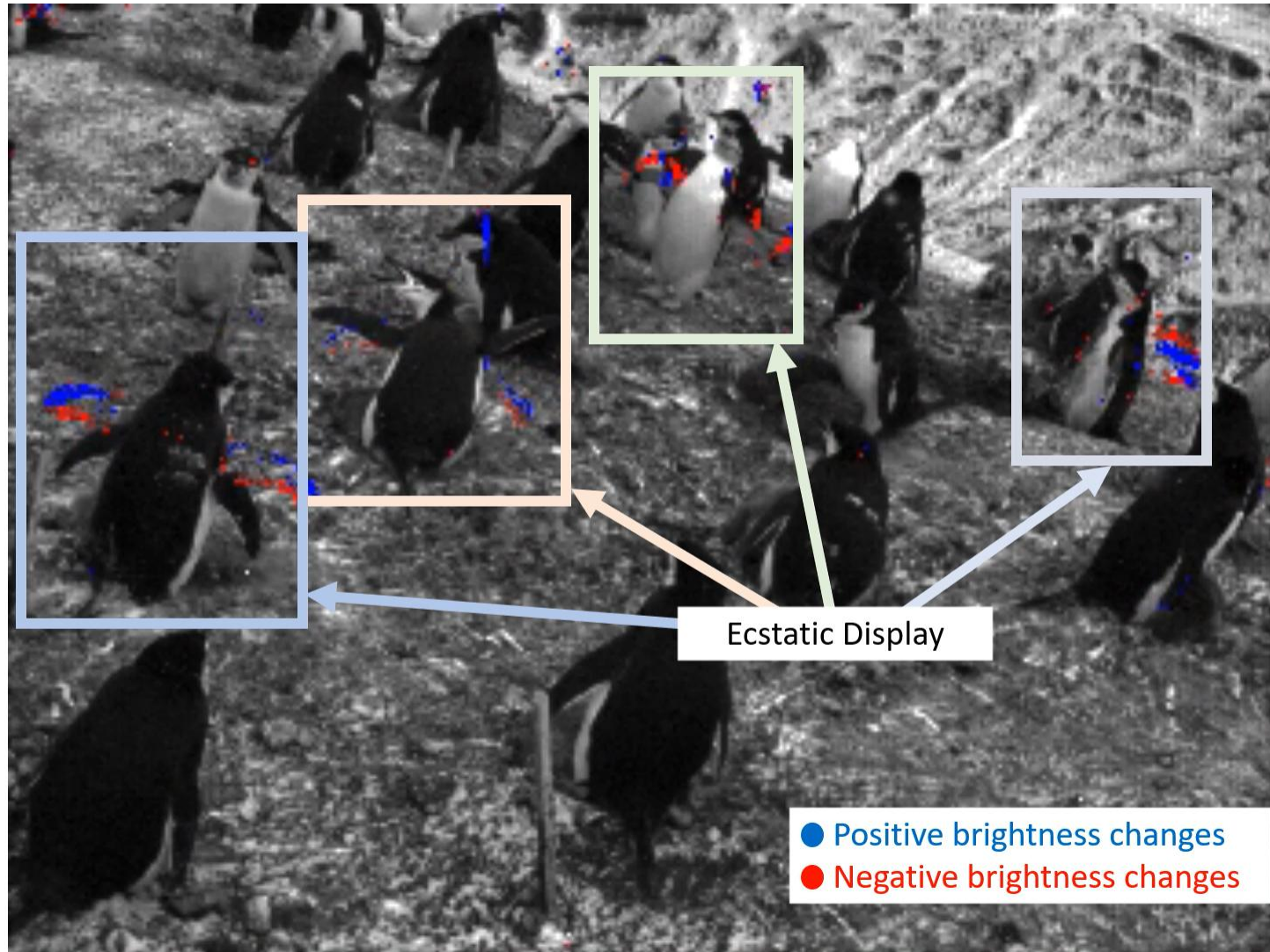


Event cameras for animal monitoring

Visualization of Ecstatic Display (ED)



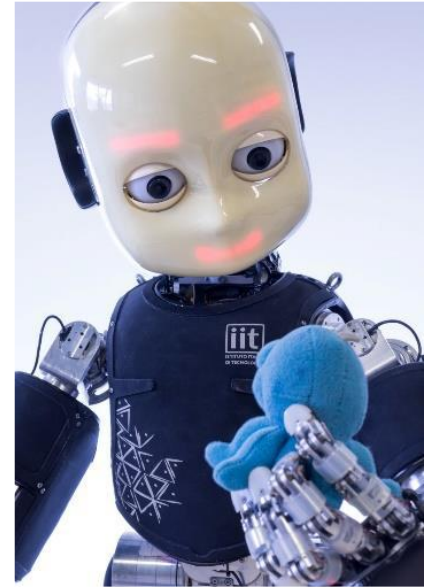
Event cameras for animal monitoring



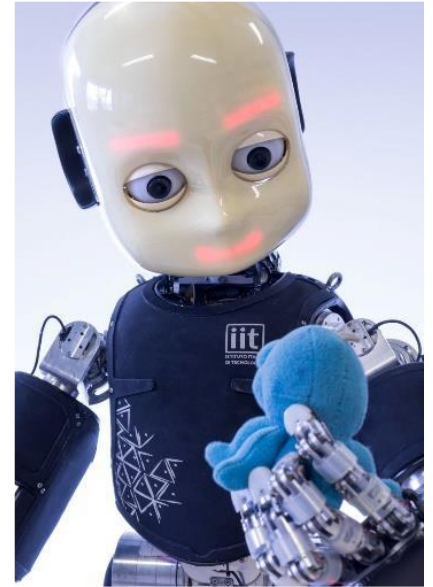
Research

Two main challenges:

- Processing the growing amount of data acquired by a robot, in an **optimal** and **efficient** way to better perceive the environment surrounding it, (i.e., designing **better perception algorithms**).
- Bringing together the **sensing** and the **control** parts of robots to adapt to dynamic environments.

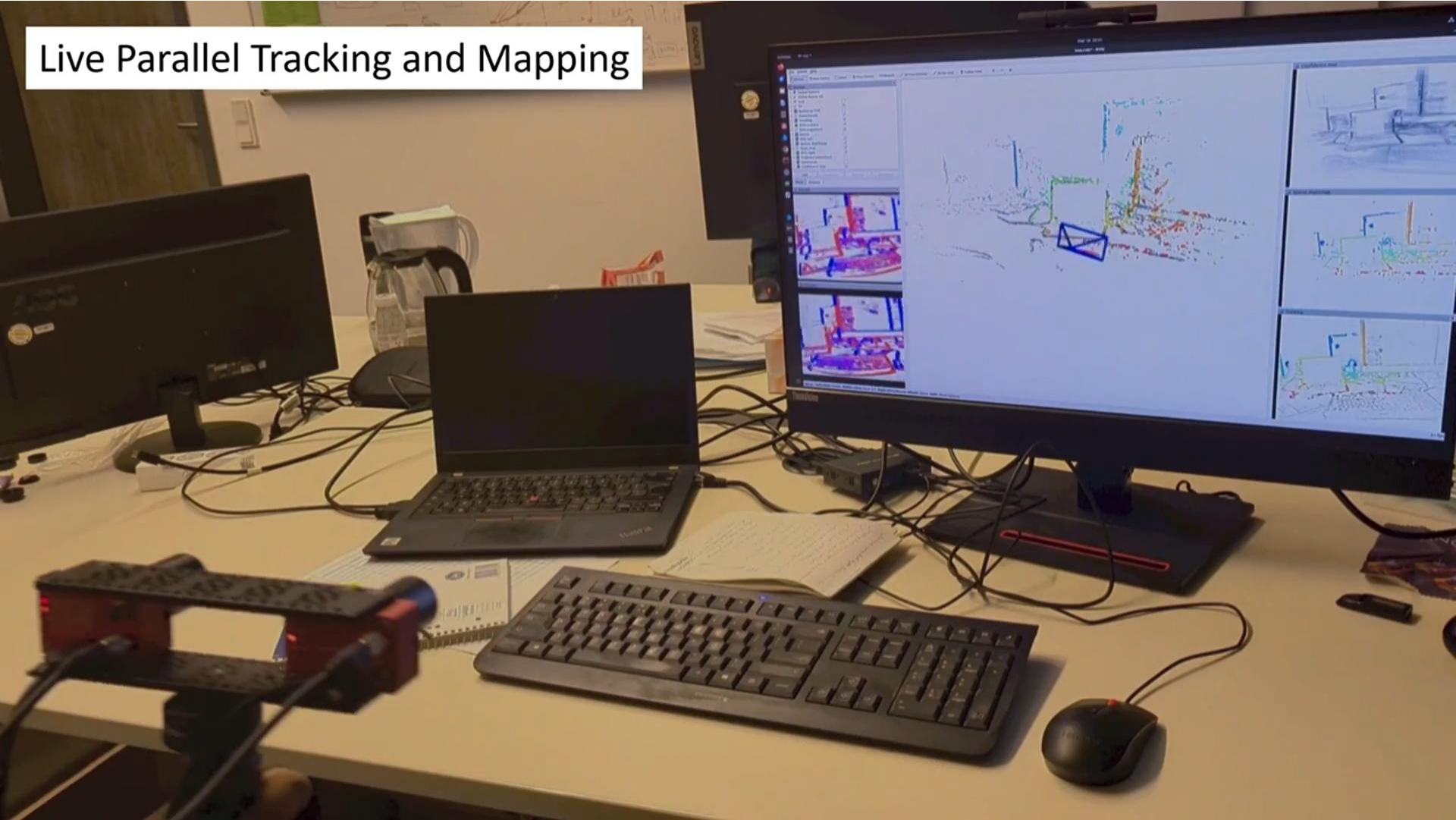


Research

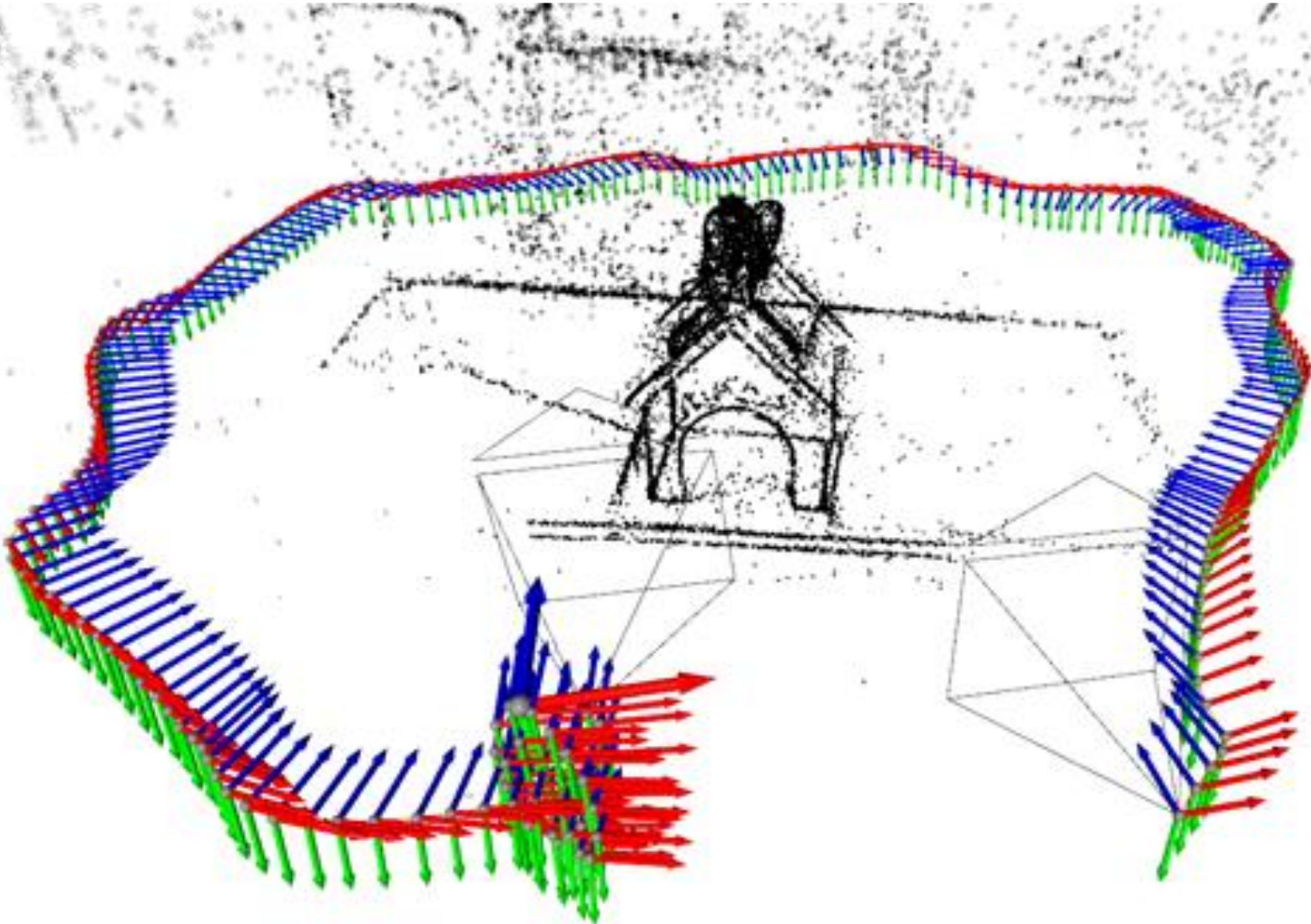


Event-based Stereo SLAM (ECCVW 2024)

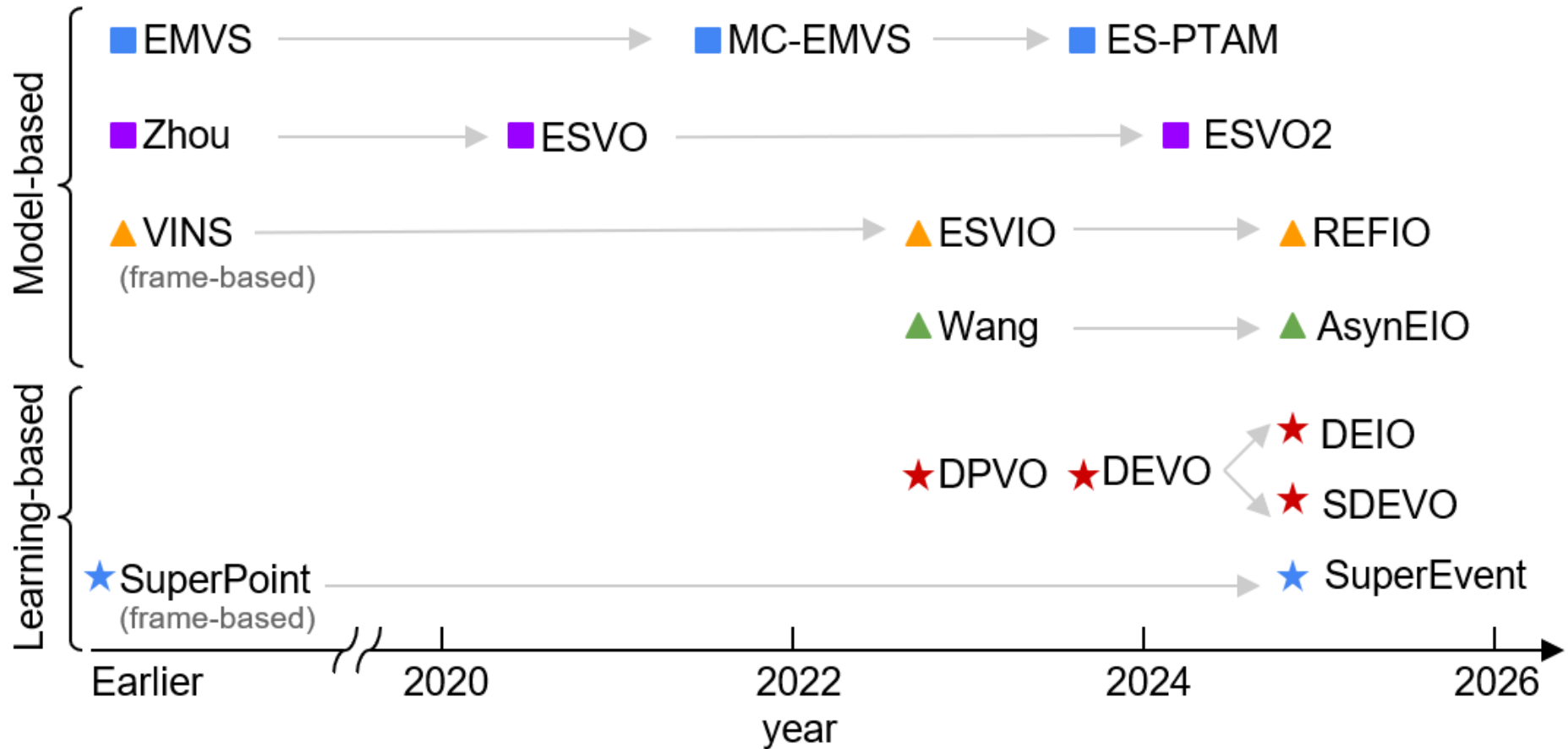
Live Parallel Tracking and Mapping



Event-based SLAM



Evolution of Stereo VO/SLAM methods



Event-Based Stereo Depth Estimation: A Survey

Suman Ghosh  and Guillermo Gallego 

Abstract—Stereopsis has widespread appeal in computer vision and robotics as it is the predominant way by which we perceive depth to navigate our 3D world. Event cameras are novel bio-inspired sensors that detect per-pixel brightness changes asynchronously, with very high temporal resolution and high dynamic range, enabling machine perception in high-speed motion and broad illumination conditions. The high temporal precision also benefits stereo matching, making disparity (depth) estimation a popular research area for event cameras ever since their inception. Over the last 30 years, the field has evolved rapidly, from low-latency, low-power circuit design to current deep learning (DL) approaches driven by the computer vision community. The bibliography is vast and difficult to navigate for non-experts due to its highly interdisciplinary nature. Past surveys have addressed distinct aspects of this topic, in the context of applications, or focusing only on a specific class of techniques, but have overlooked stereo datasets. This survey provides a comprehensive overview, covering both instantaneous stereo and long-term methods suitable for simultaneous localization and mapping (SLAM), along with theoretical and empirical comparisons. It is the first to extensively

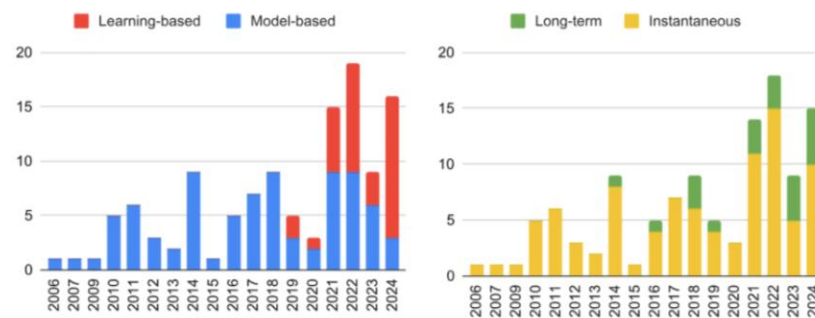
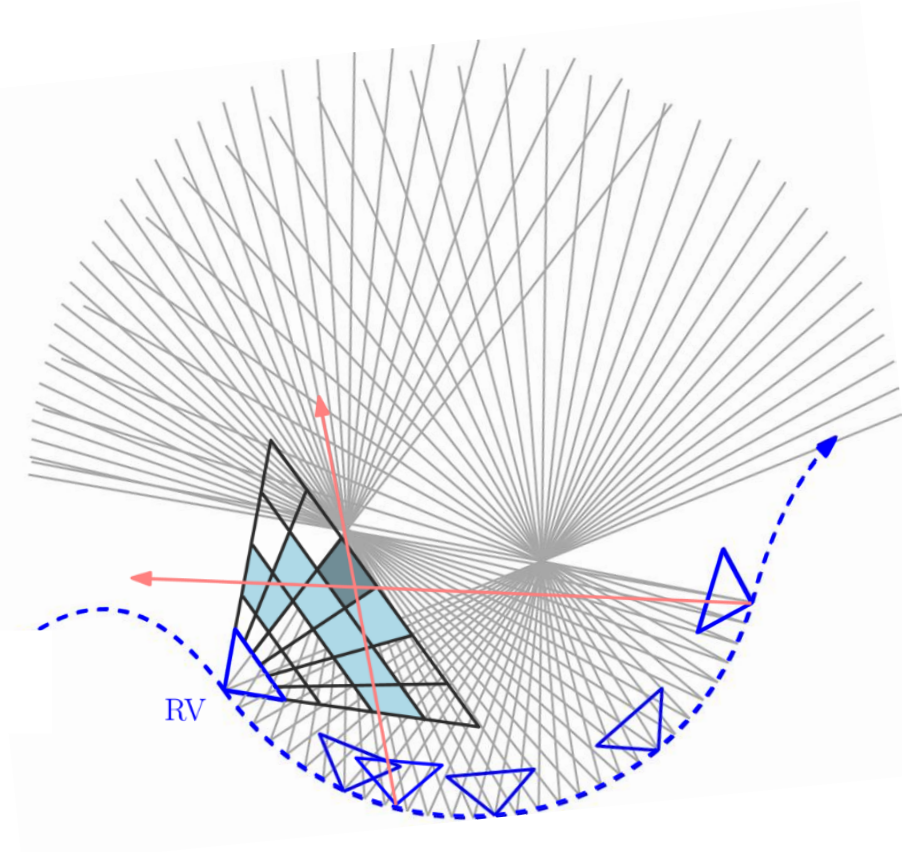


Fig. 1. Publications on event-based stereo depth estimation in the last two decades, classified according to criteria #1: whether they are model-based (i.e., hand-crafted) or learning-based (i.e., data-driven) methods (left), and #2: whether they produce instantaneous depth outputs or have long-term motion consistency (right). Plots created from a compiled spreadsheet of growing number of papers.

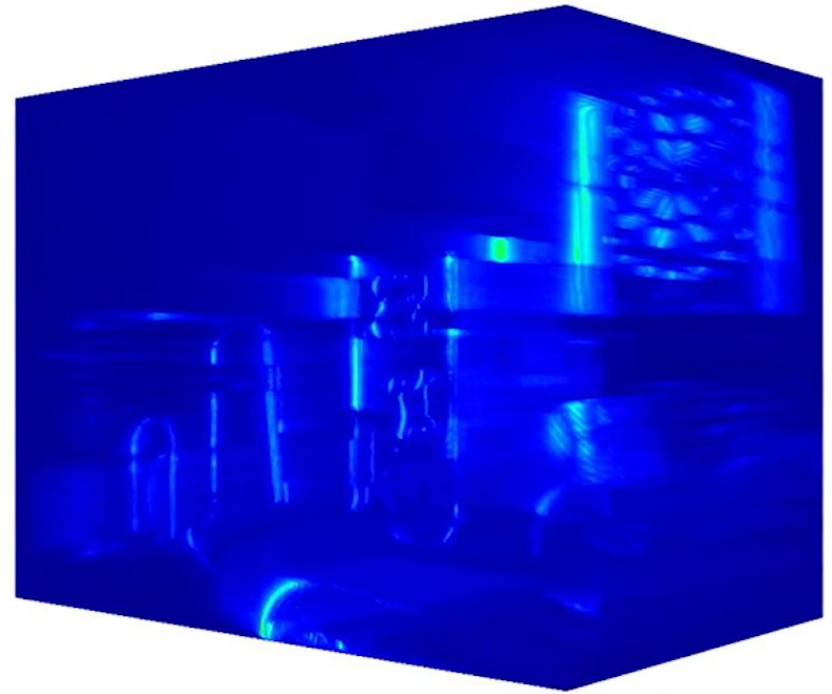
interest in computer vision and robotics because it tries to mimic the same functionality of the human brain (i.e., inverting

3D Reconstruction



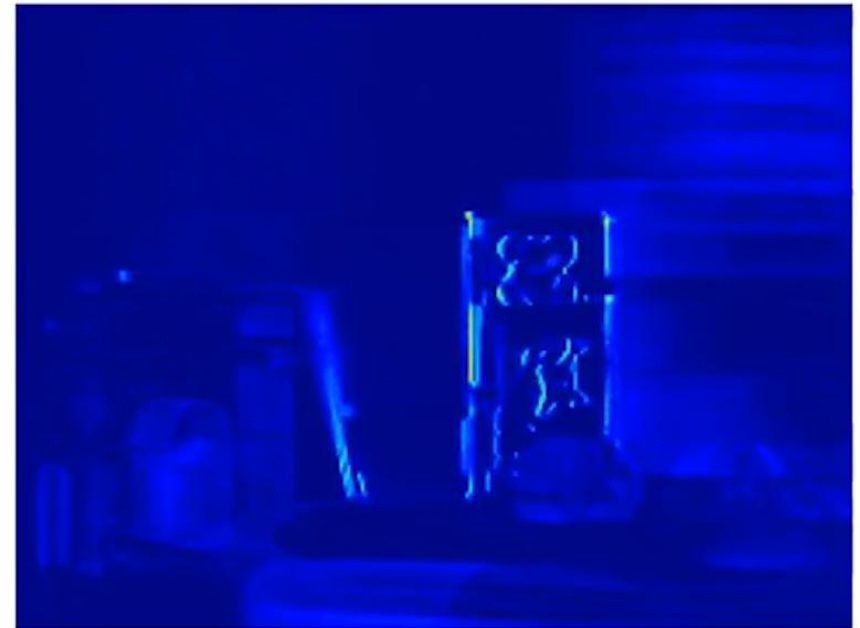
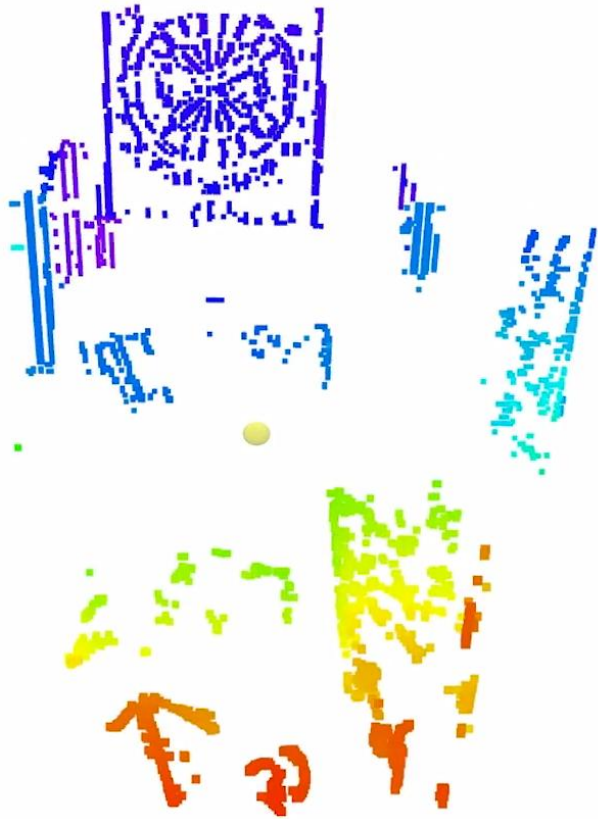
Moving event camera
with known poses

Back-project events into space



Ray-density
Disparity space image (DSI)

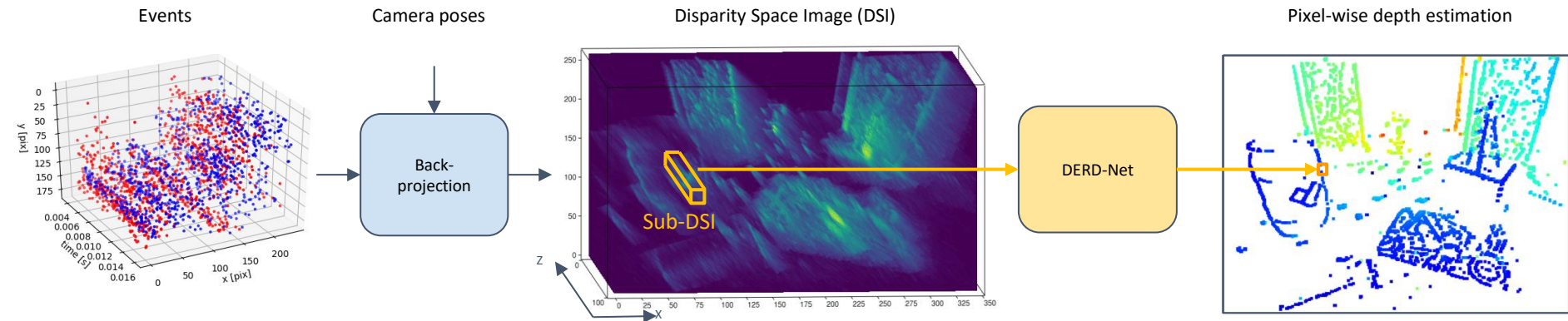
3D Reconstruction



Ray-density
Disparity space image (DSI)

Can we do better than argmax (counting rays)?

DERD-Net: Learning Depth from DSIs

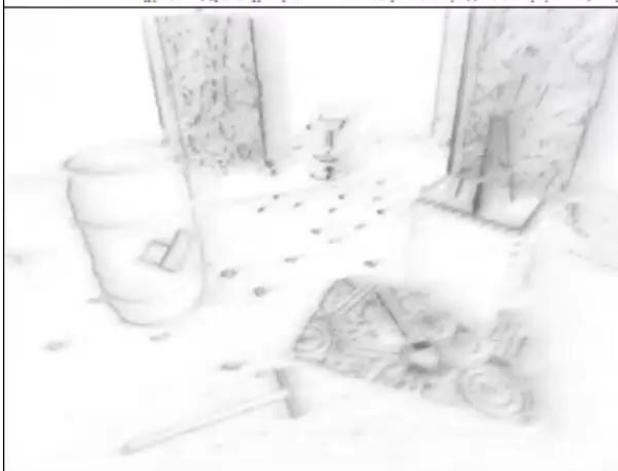
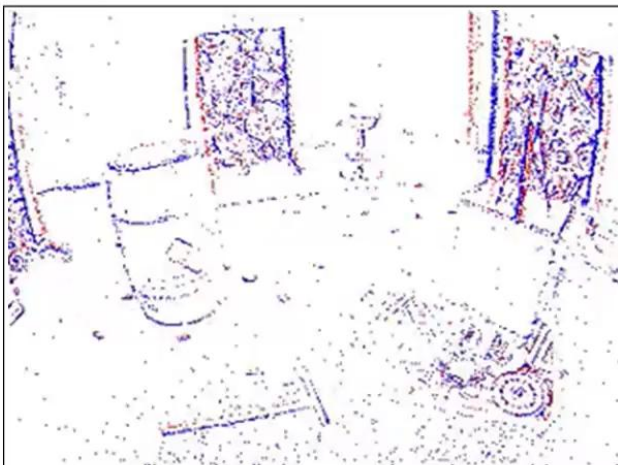


Inference time: 0.37 ms

Model size: < 1 MB

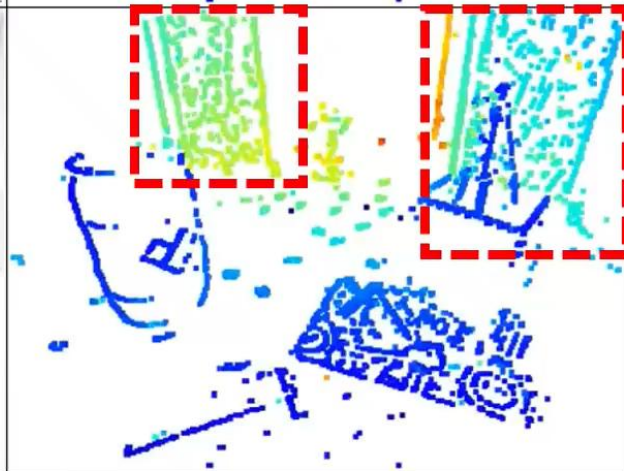
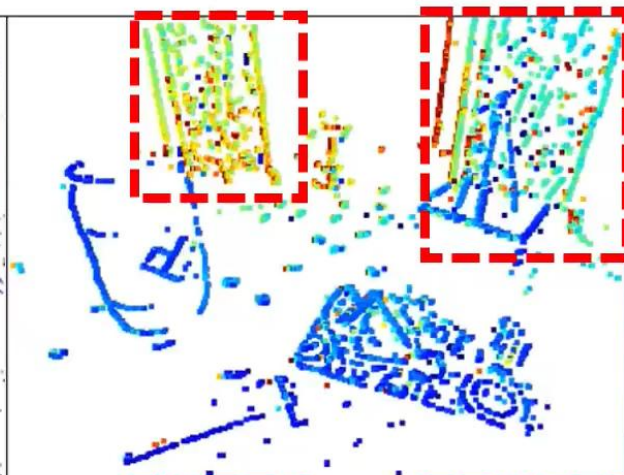
Results

Events



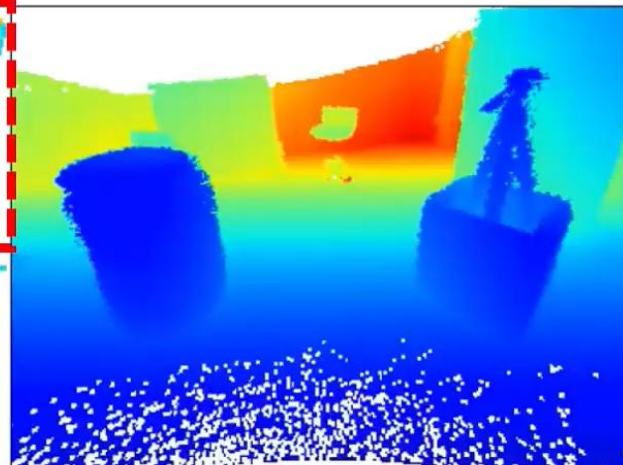
Confidence Map (from DSI)

MC-EMVS Denser Filter



Ours (DERD-Net)

Ground Truth



Masked Ground Truth

EventHub: Data Factory for Generalizable Event-Based Stereo Networks without Active Sensors

CVPR 2026

Luca Bartolomei^{1,2,3}

Fabio Tosi²

Matteo Poggi^{1,2}

Stefano Mattoccia^{1,2}

Guillermo Gallego³

¹Advanced Research Center on Electronic System (ARCES), University of Bologna, Italy

²Department of Computer Science and Engineering (DISI), University of Bologna, Italy

³TU Berlin, Robotics Institute, Einstein Center Digital Future, SCIOI Excellence Cluster, Germany



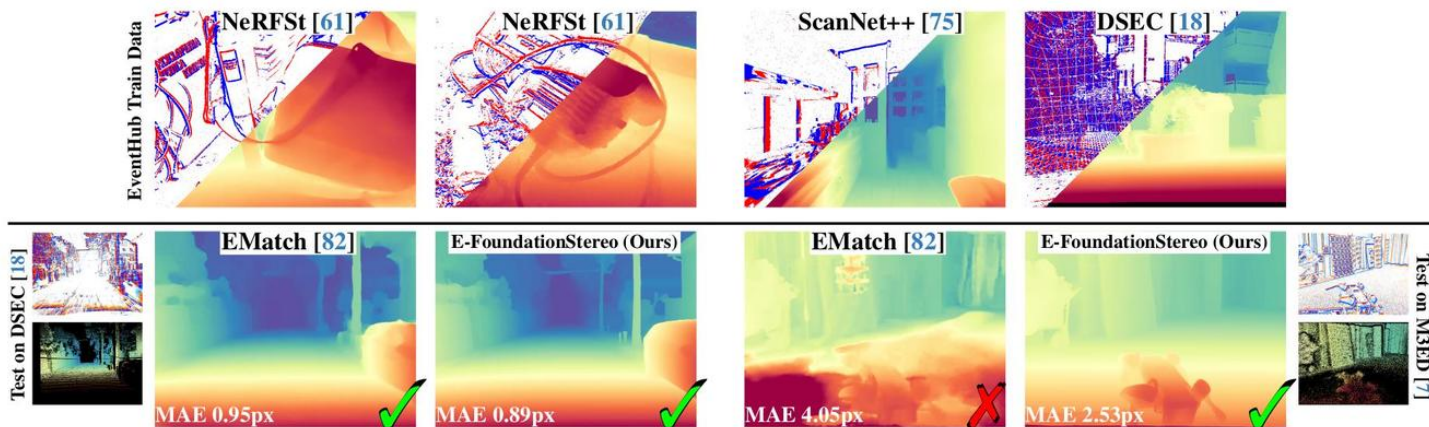
Paper



Poster



Code

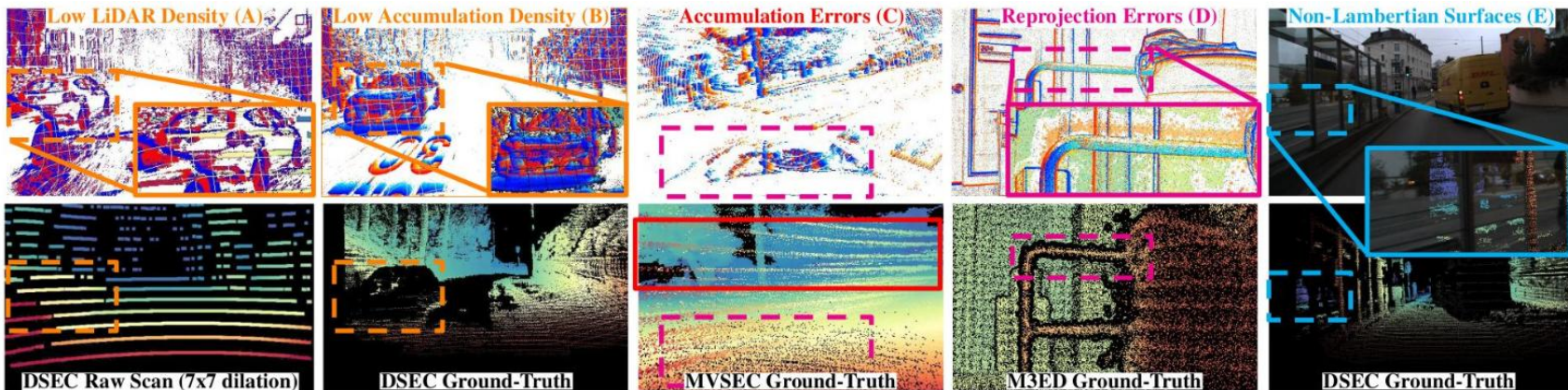


EventHub Overview. Our framework combines novel view synthesis (SVRaster) for synthetic data generation with cross-modal distillation from RGB stereo foundation models to create high-quality training data for event-based stereo networks without ground-truth LiDAR annotations. The resulting data enables training of state-of-the-art, highly generalizable event stereo models.

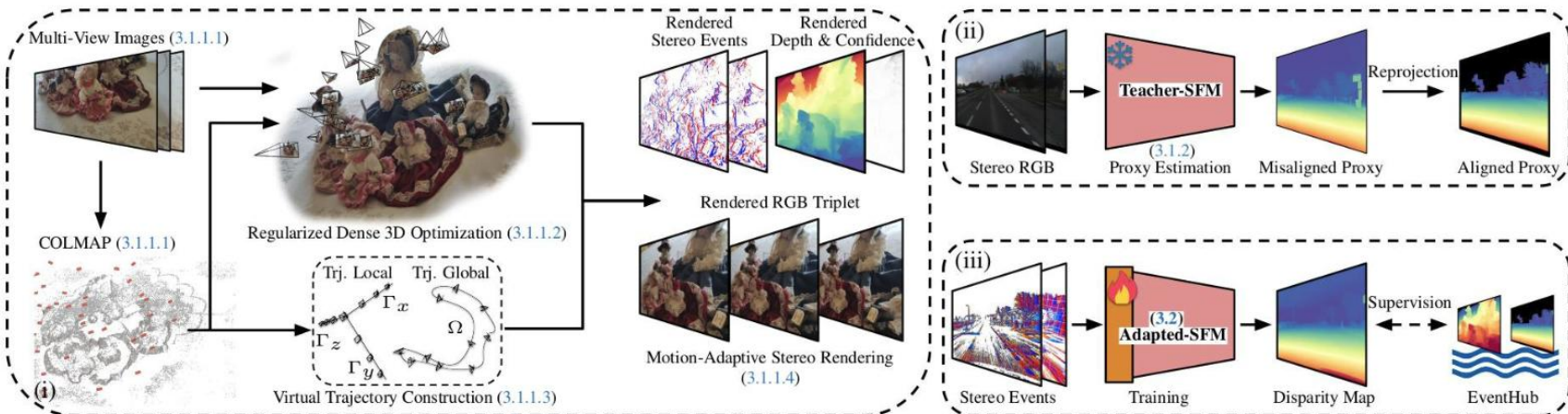
Code: <https://bartn8.github.io/eventhub/>

Video: <https://youtu.be/XD-hpmiXXT0>

EventHub (ICCV 2025)



Limitations of LiDAR-supervised real-world datasets. Despite their popularity, LiDAR annotations remain sparse (A), poorly capture dynamic scenes (B-C), are prone to reprojection errors (D), and struggle on transparent or reflective surfaces (E).



EventHub Framework Overview. Our framework employs two complementary approaches: (i) **Event Data Factory**: [SVRaster](#) generates synthetic event stereo pairs and depth labels from sparse RGB images via virtual camera trajectories; (ii) **Stereo Cross-Modal Distillation**: existing RGB stereo models produce proxy depth labels for real event data in calibrated RGB-Event stereo setups; (iii) Both data sources are combined in EventHub to train/adapt event stereo networks.

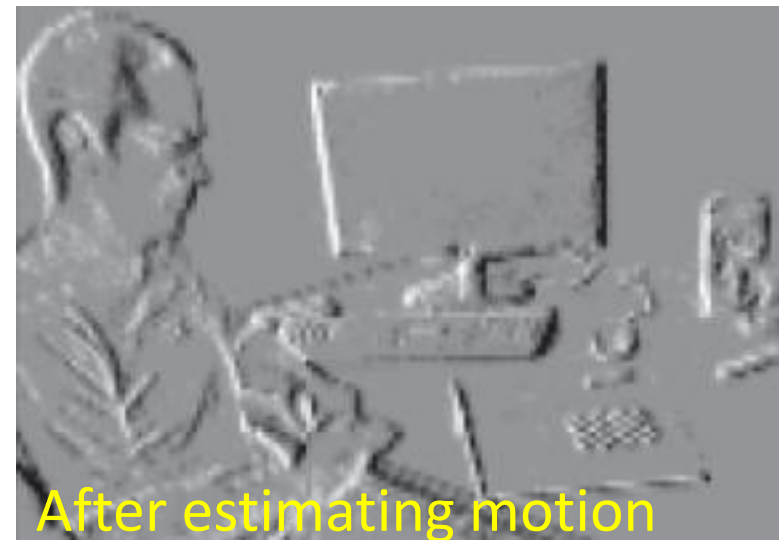
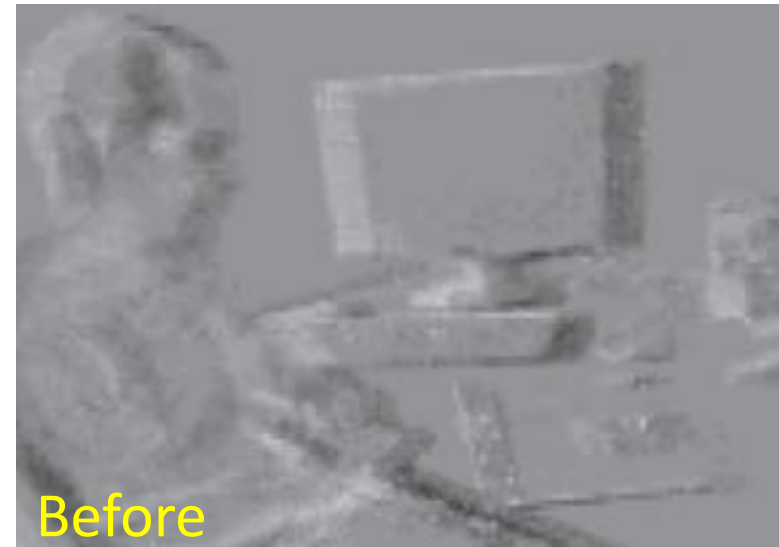
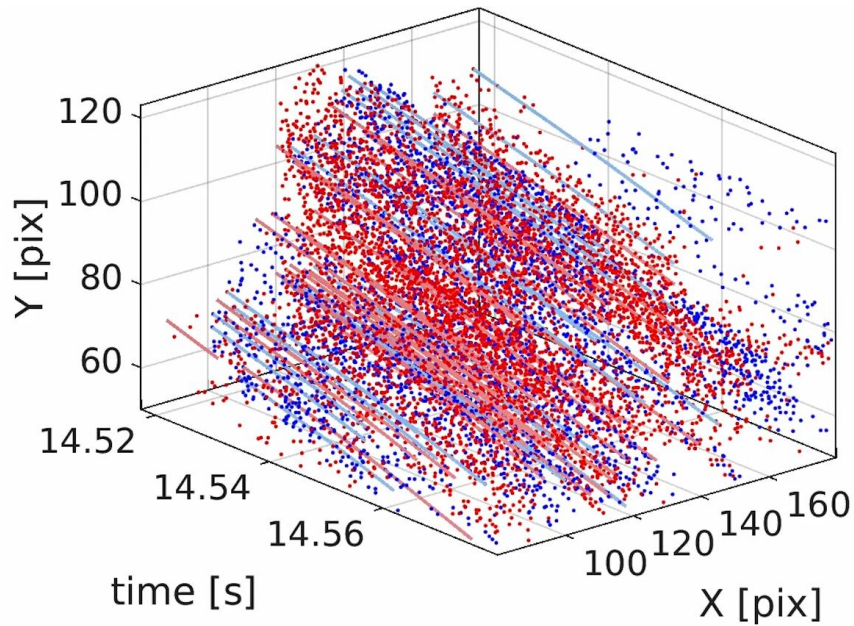
Events respond to motion...

Contrast / Focus Maximization Framework

A **framework** that allows to solve multiple computer vision problems with **event cameras** in a **unified** way:

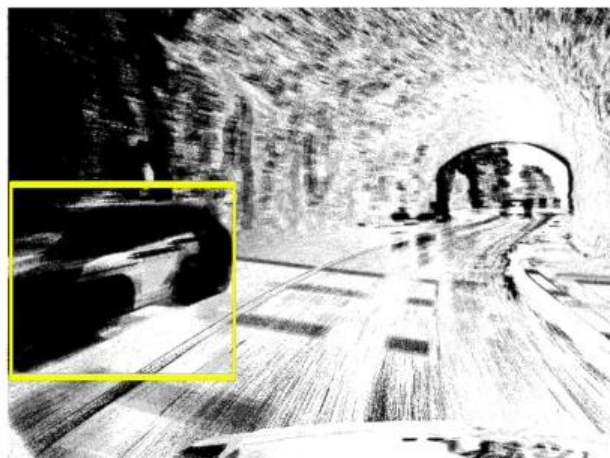
- **Optical flow estimation**
- **3D reconstruction**
- **Ego-motion estimation**
- **Homographic motion estimation**
- **Motion segmentation**
- **Non-linear point-trajectory estimation**
- **... (many more)**

Contrast Maximization Framework



Dense Optical Flow (ECCV 2022, TPAMI 2024)

**Independently
moving object
(IMO)**



Input: Events



Output: Warped events



Reference frame (not used)

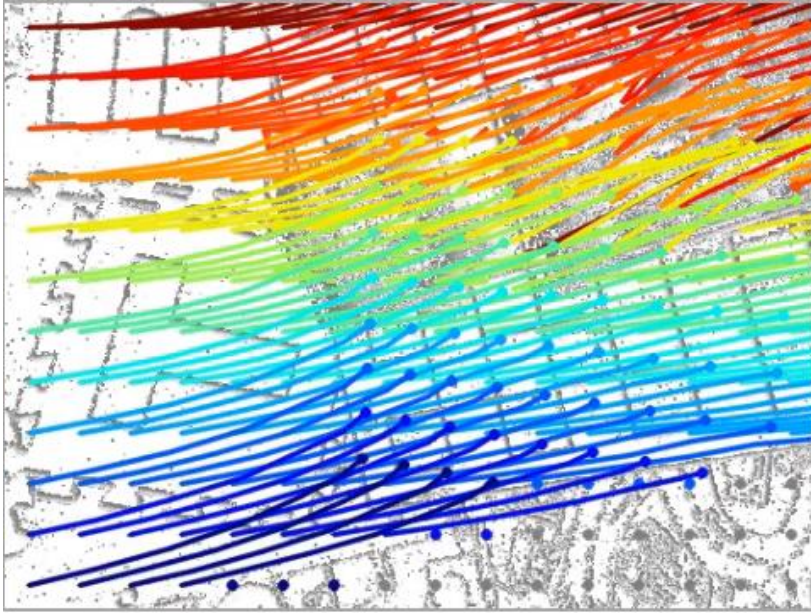


Output: Estimated flow

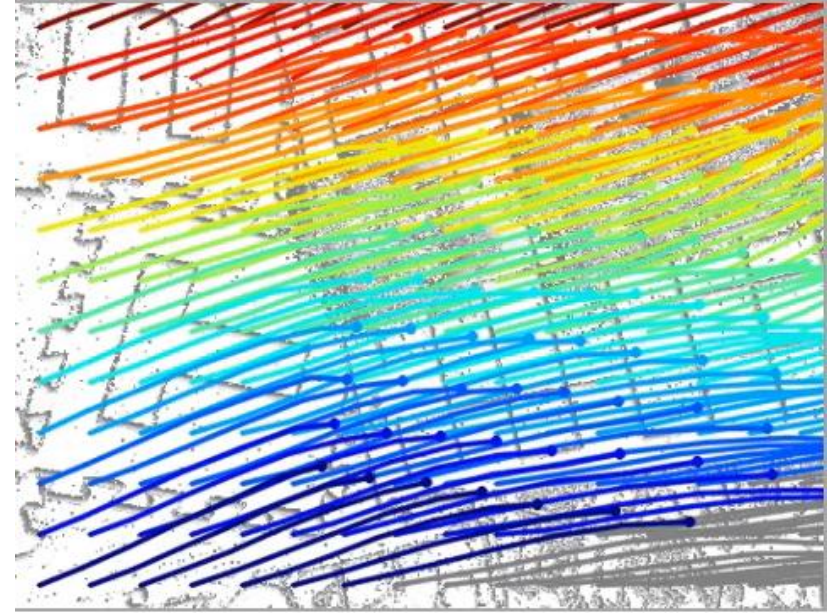


Code available: https://github.com/tub-rip/event_based_optical_flow

Trajectory Prediction (ECCV 2024)

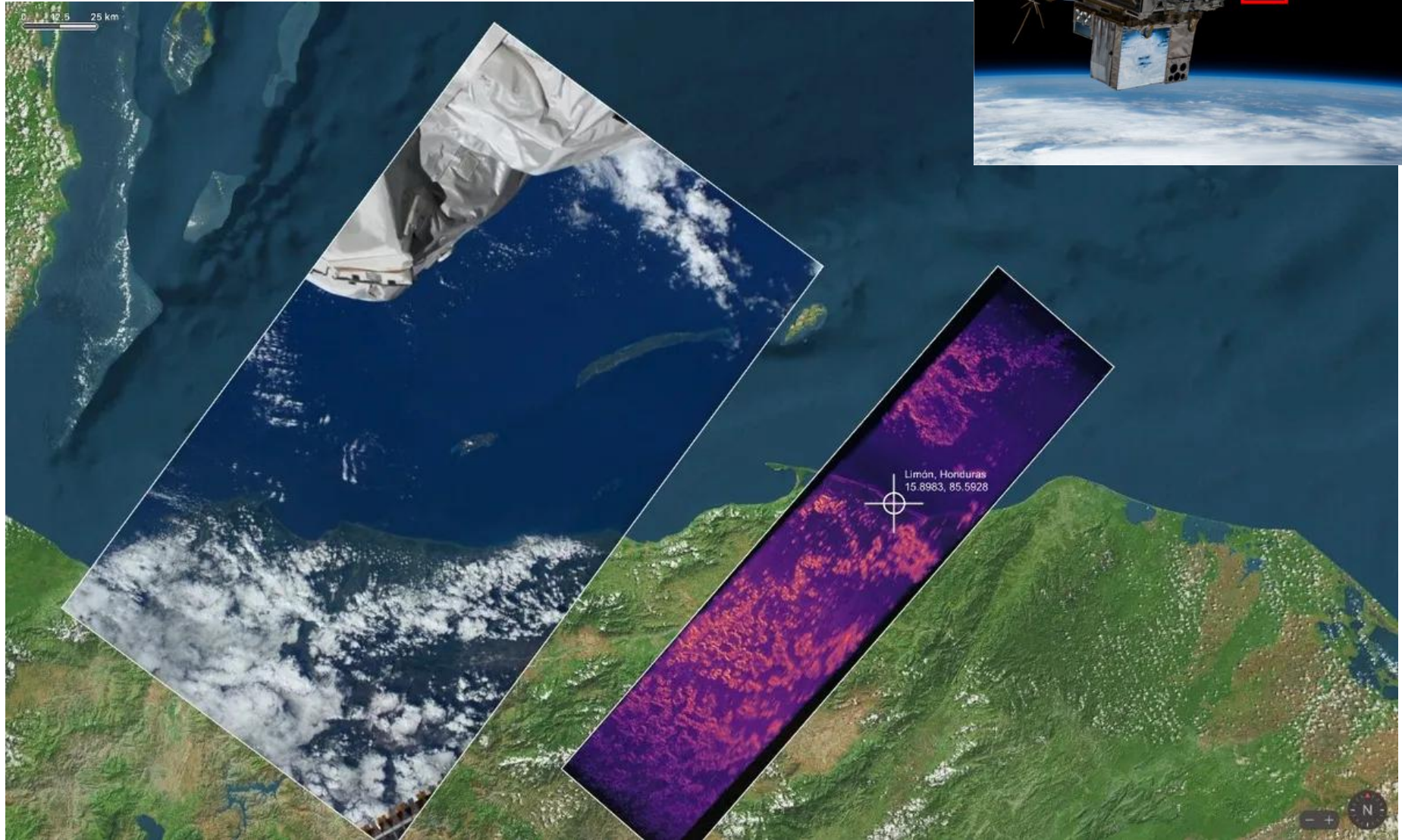


GT



Ours

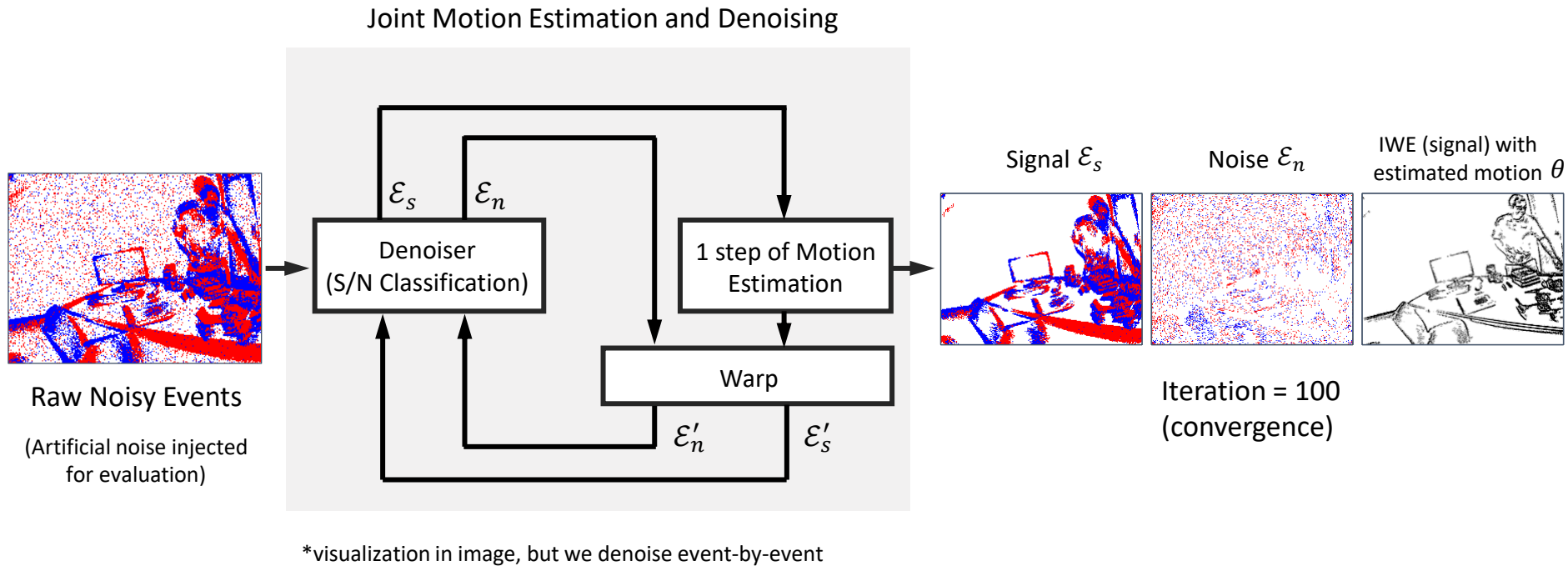
Earth observation from orbit



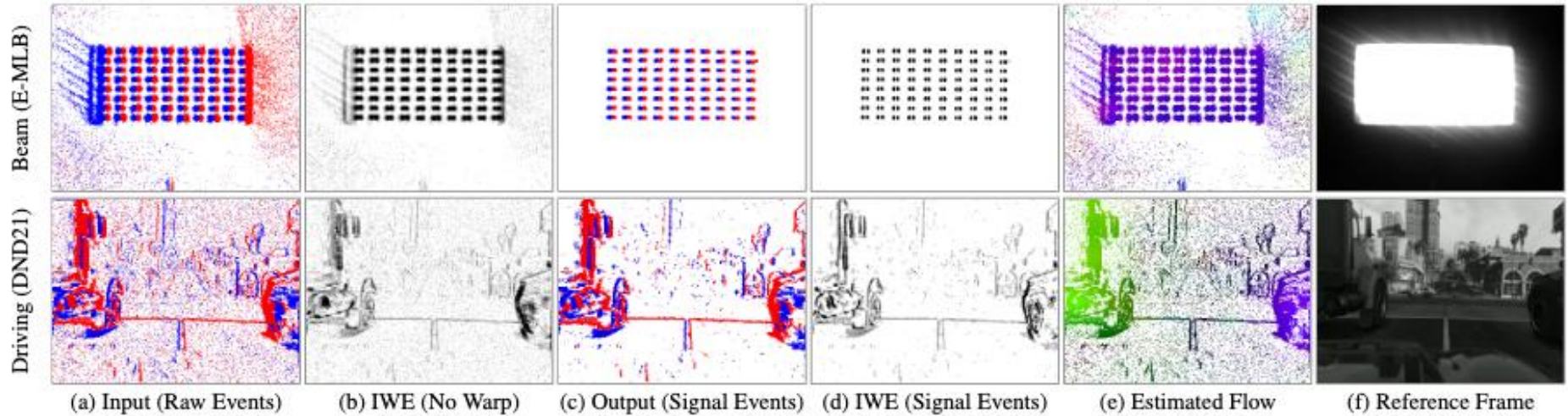
ESMD

ICCV 2025

Joint Motion Estimation and Denoising



Results | Denoising (evaluation without GT)



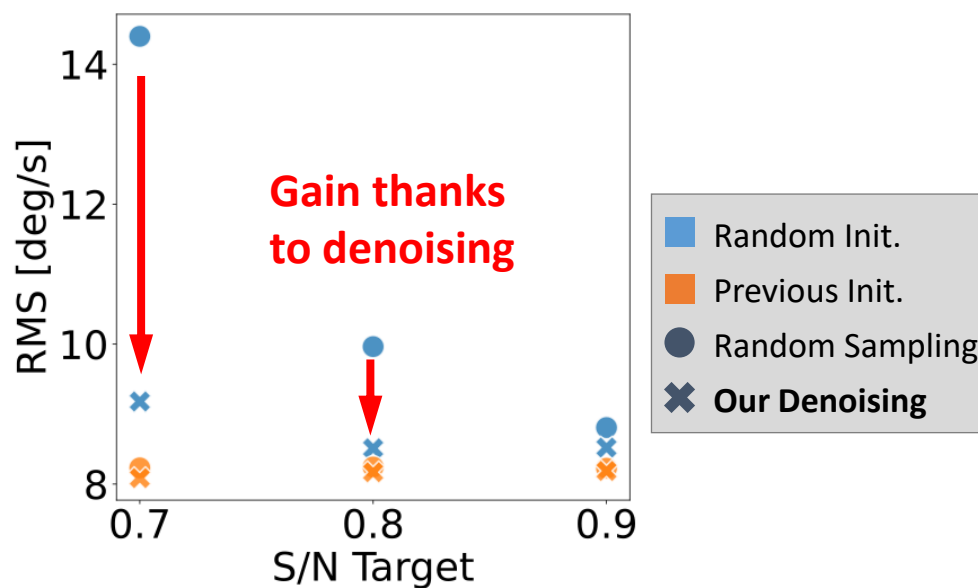
		E-MLB (Day)				E-MLB (Night)				DND21
		ND1	ND4	ND16	ND64	ND1	ND4	ND16	ND64	
Model-based	Raw	0.821	0.824	0.815	0.786	0.890	0.824	0.786	0.768	0.869
	BAF [8]	0.861	0.869	0.876	0.890	0.946	<u>0.973</u>	0.992	0.942	0.920
	KNoise [29]	0.846	0.837	0.830	0.807	0.954	0.956	0.871	0.817	0.887
	DWF [23]	0.878	0.876	0.866	0.865	0.923	0.962	0.988	0.932	0.905
	EvFlow [49]	0.848	0.878	0.868	0.833	0.969	0.983	<u>0.889</u>	0.797	1.006
	Ynoise [13]	0.866	0.863	0.857	0.821	1.009	0.943	0.875	0.792	0.966
	TS [31]	0.877	0.887	0.870	0.837	1.033	0.944	0.886	0.797	0.985
	IETS [5]	0.772	0.785	0.777	0.753	0.950	0.823	0.804	0.711	0.900
	GEF [10]	1.051	<u>0.938</u>	<u>0.935</u>	<u>0.927</u>	<u>1.027</u>	0.955	0.946	<u>0.935</u>	0.932
	Ours	<u>0.938</u>	0.958	0.986	0.950	1.037	0.961	0.945	0.932	<u>0.992</u>
Learning	EventZoom [11]	0.996	0.988	0.996	0.970	1.055	1.007	1.010	0.988	1.059
	EDnCNN [4]	0.887	0.908	0.903	0.912	1.001	1.024	1.079	<u>1.086</u>	0.977
	MLPF [23]	0.851	0.855	0.846	0.840	0.926	0.928	0.910	0.906	0.944
	EDformer [27]	<u>0.952</u>	<u>0.955</u>	<u>0.956</u>	<u>0.942</u>	<u>1.048</u>	<u>1.019</u>	<u>1.076</u>	1.099	<u>1.041</u>

Our method:

- Ranks 1st or 2nd among model-based methods.
- Achieves competitive scores even with learning-based methods.

Results | Motion Estimation Improvement

Ego-motion



Optical flow

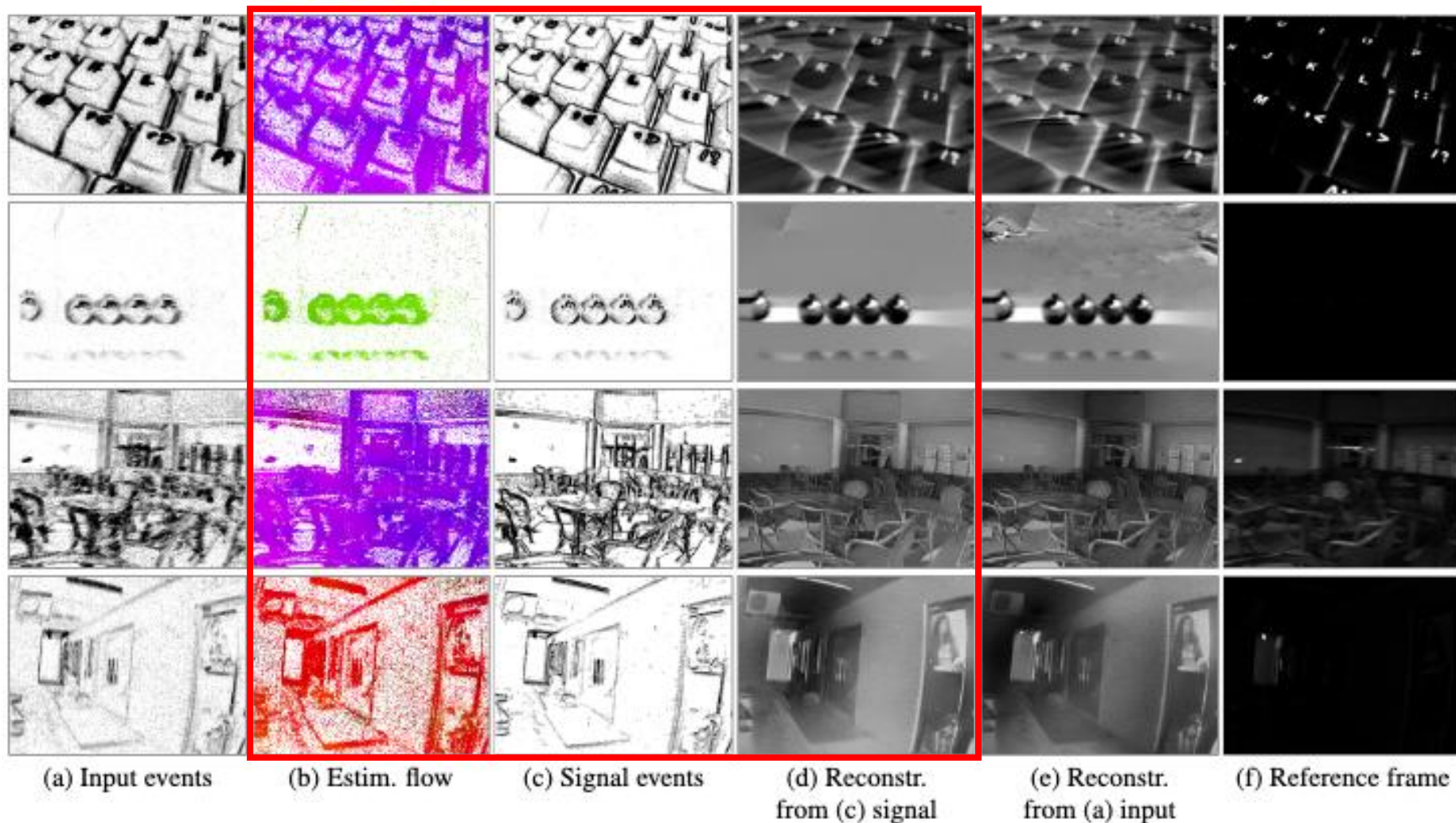
	indoor1		indoor2	
	EPE ↓	%Out ↓	EPE ↓	%Out ↓
MultiCM [47]	2.676	32.098	2.746	33.474
Ours	2.517	29.475	2.695	33.047

	indoor3		outdoor1	
	EPE ↓	%Out ↓	EPE ↓	%Out ↓
MultiCM [47]	2.732	33.219	1.960	18.160
Ours	2.654	32.241	1.865	17.554

Gain thanks
to denoising

Our method improves motion estimation thanks to the joint denoising.

Results | Image Reconstruction Improvement



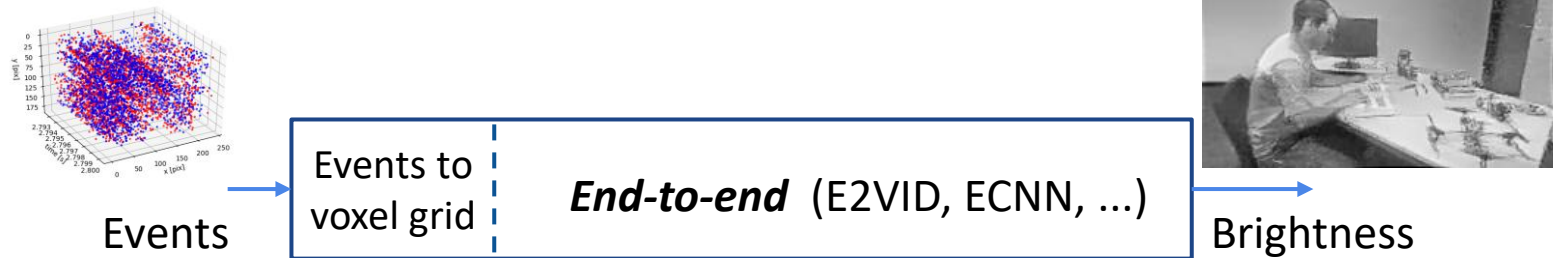
**Our method achieves less artifacts and better quality
thanks to the denoising**

E2FAI

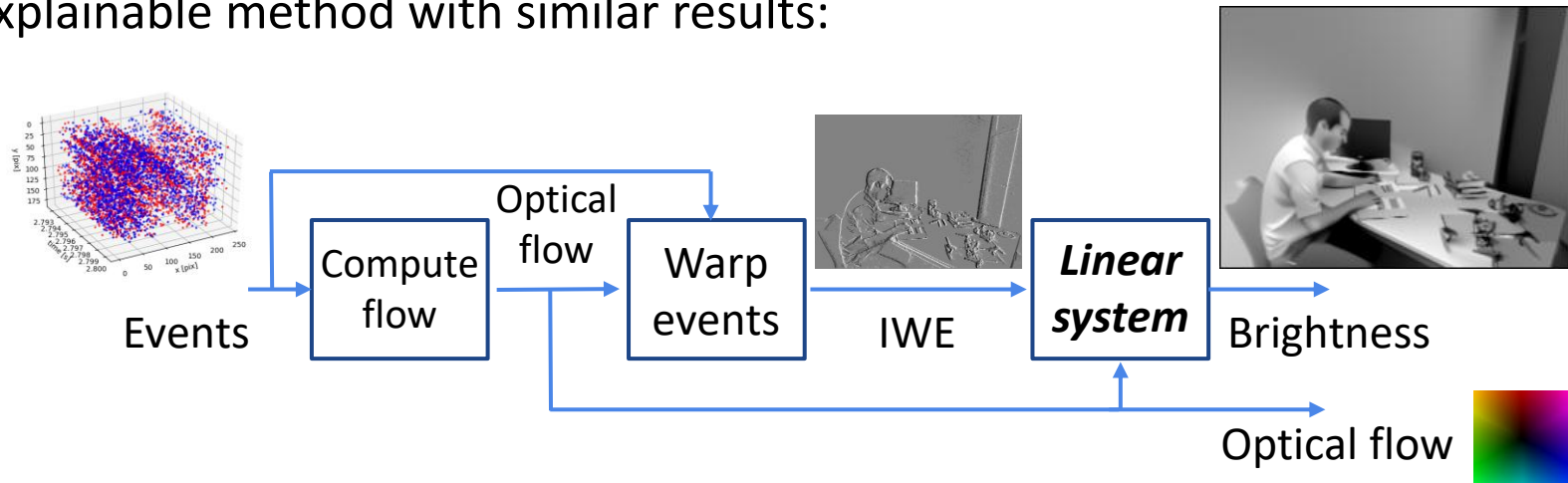
ICCV 2025

Image reconstruction

State of the art: Recurrent Neural Network (black-box)

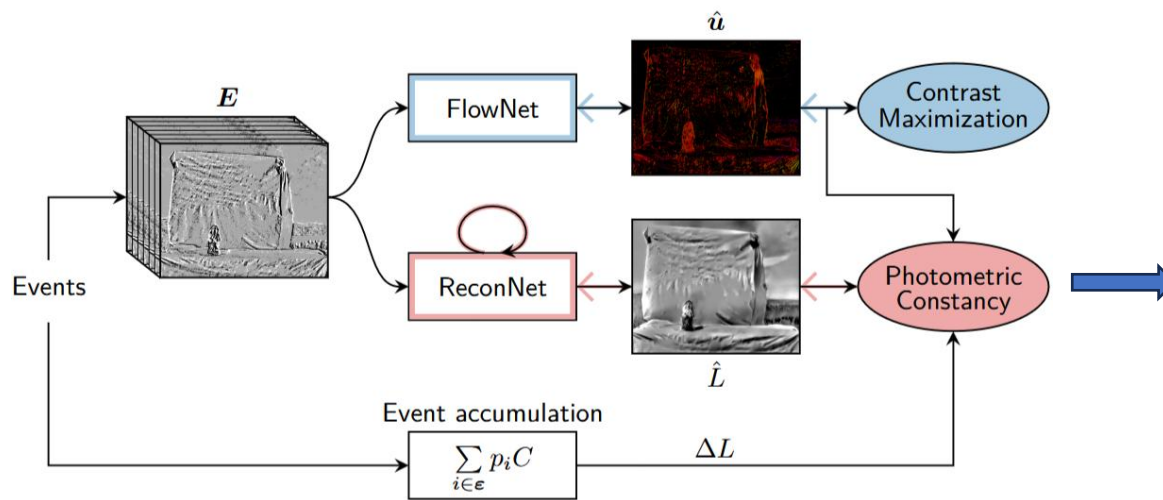


An explainable method with similar results:



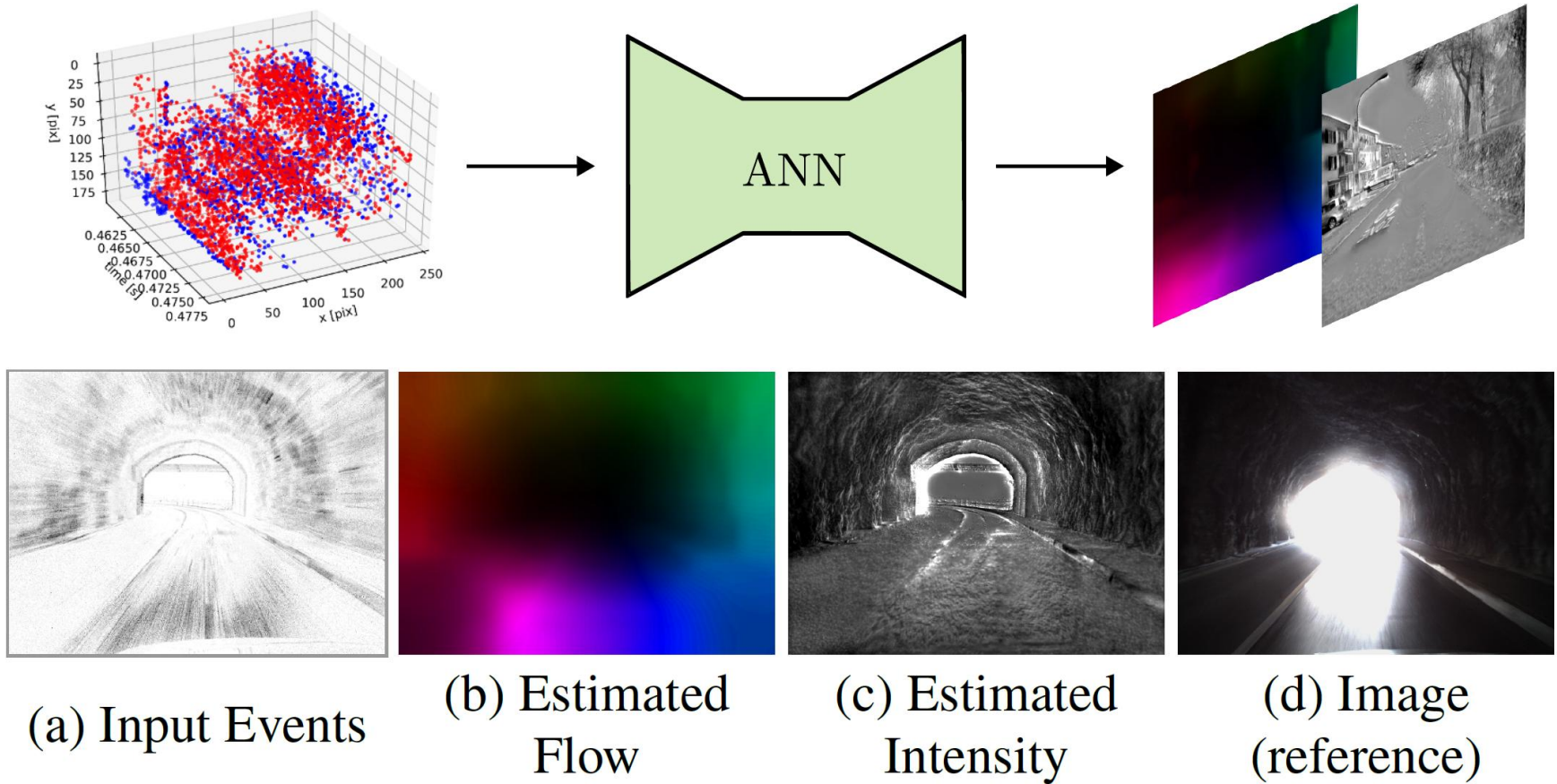
Sequential Learning of Flow and Intensity

- First train the optical flow DNN (CMax)
- Then train the image reconstruction DNN, unsupervised
- Linearized Event Generation Model



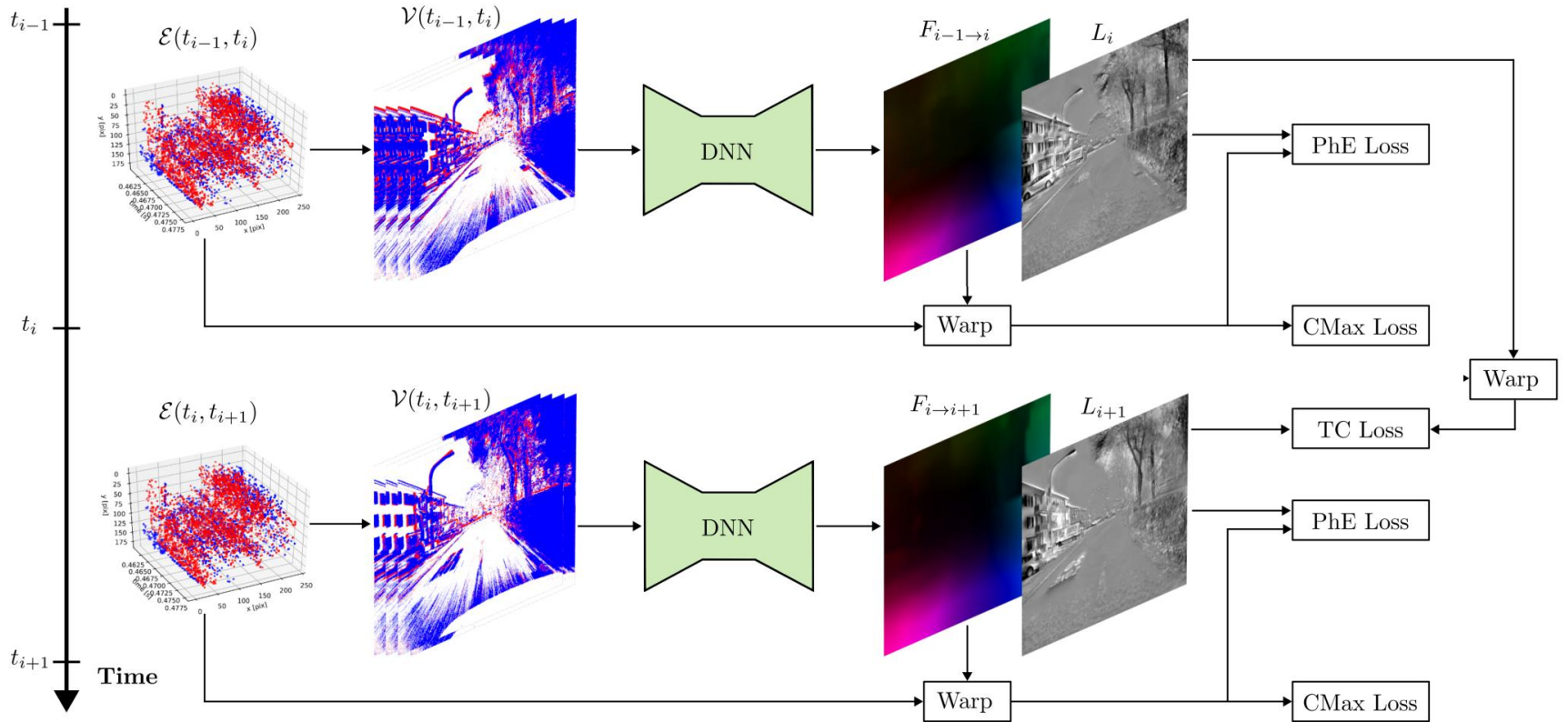
E2FAI: Events to Flow and Intensity

- Jointly estimating both, exploiting their synergies.



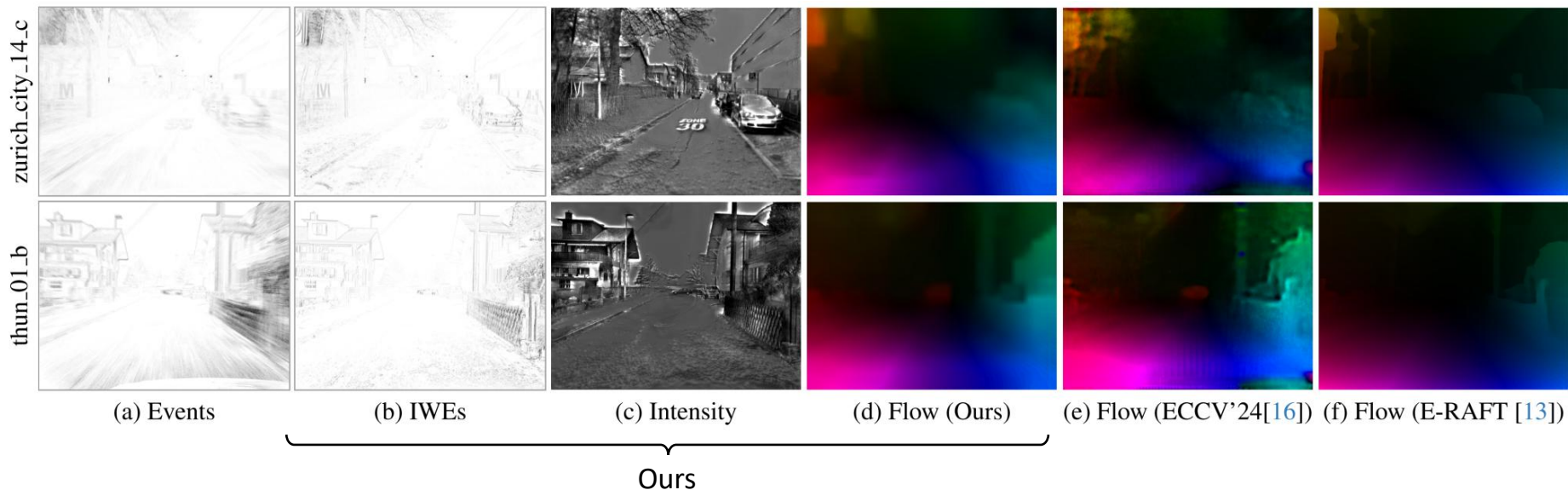
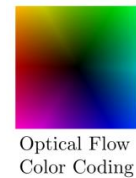
E2FAI: Events to Flow and Intensity

- Jointly estimating both, exploiting their synergies.



$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{PhE}} + \lambda_2 \mathcal{L}_{\text{CMax}} + \lambda_3 \mathcal{L}_{\text{FTV}} + \lambda_4 \mathcal{L}_{\text{ITV}} + \lambda_5 \mathcal{L}_{\text{TC}}$$

E2FAI: Events to Flow and Intensity

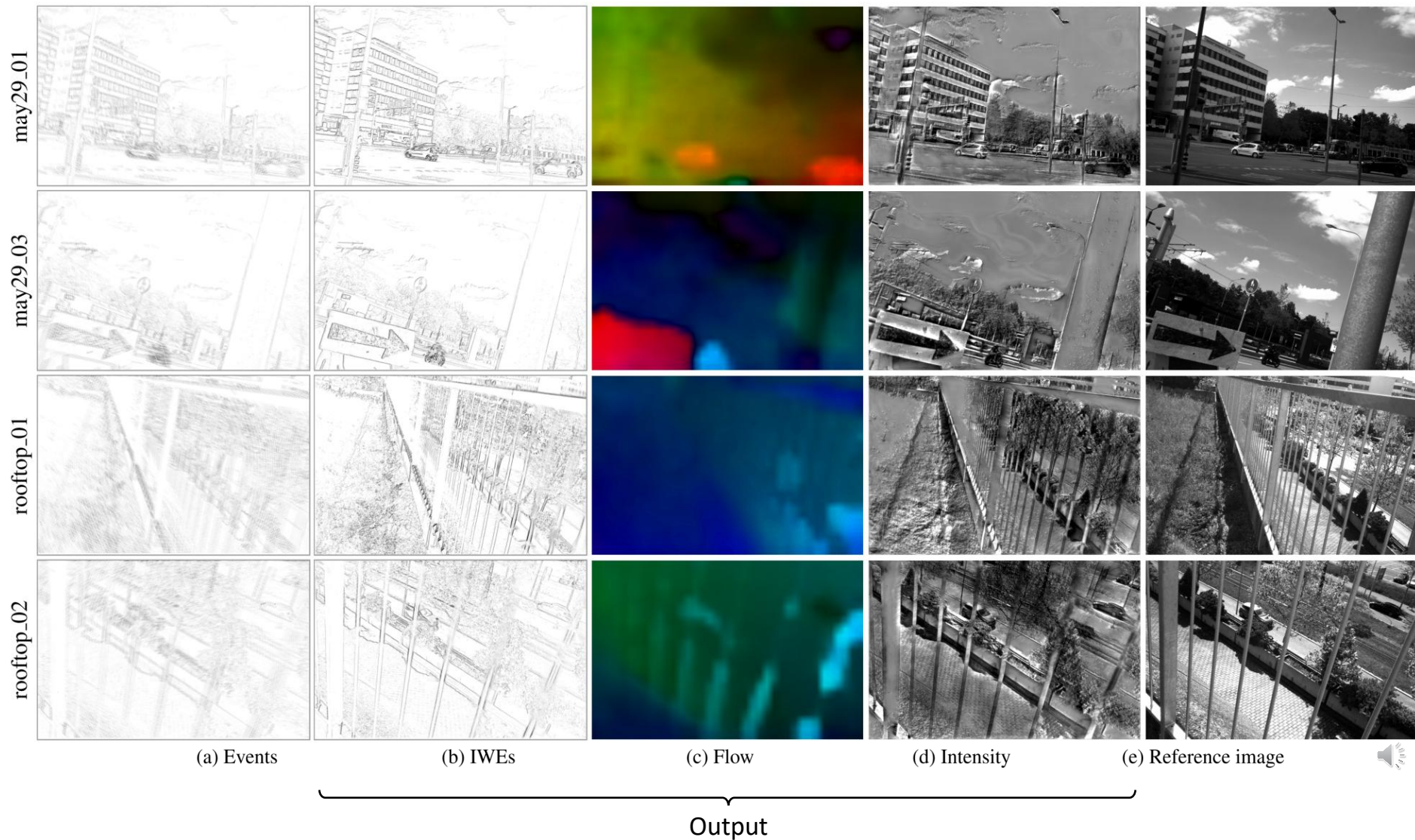


Joint estimation of flow and intensity with:

- **Best optical flow** accuracy among MB/USL (no ground truth labels required).
- Shortest **inference time**.
- **Fine-detail** intensity reconstruction.

Type	Method	$t_{\text{inf}}[\text{ms}]$	All			
			EPE ↓	AE ↓	%Out ↓	FWL ↑
SL	E-RAFT [13]	46.33	0.79	10.56	2.68	1.29
	IDNet [43]		0.72	2.72	2.04	–
MB/ USL	RTEF [3]		4.88	–	41.95	2.51
	MultiCM [36]	$9.9 \cdot 10^3$	3.47	13.98	30.86	1.37
	Paredes et al. [28]	40.1	2.33	10.56	17.77	–
	EV-FlowNet [50]		3.86	–	31.45	1.30
	MotionPriorCM [15]	17.86	3.20	8.53	15.21	1.46
	VSA-SM [45]		2.22	8.86	16.83	–
	Ours	15.12	1.78	6.44	11.24	1.79

Results on 1Mpixel data (BS-ERGB data)



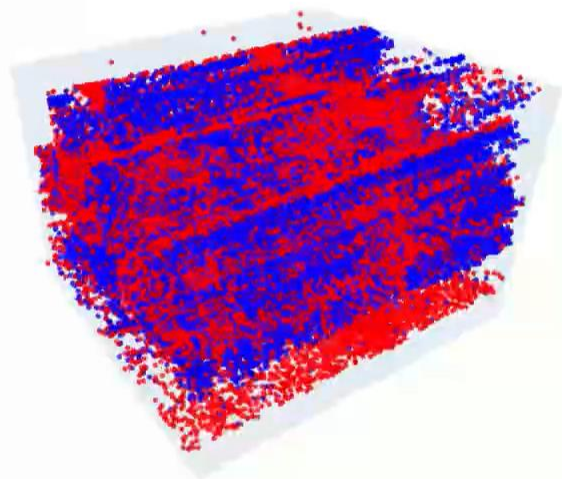
GPert

CVPR 2026

Joint Intensity, Depth and Flow estimation

- exploiting their synergies, for novel view synthesis

Visualization



Input events



Event-by-event depth



Event-by-event flow

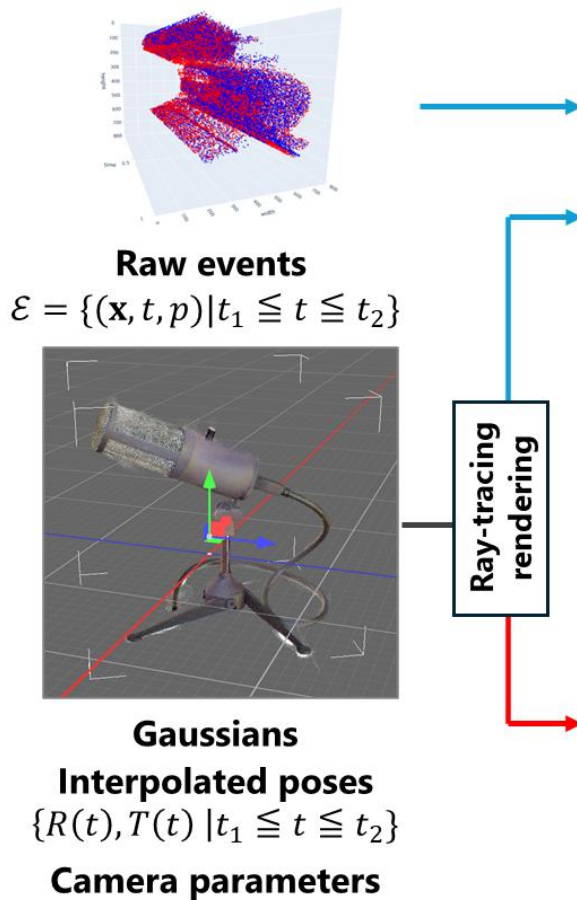


Rendered image

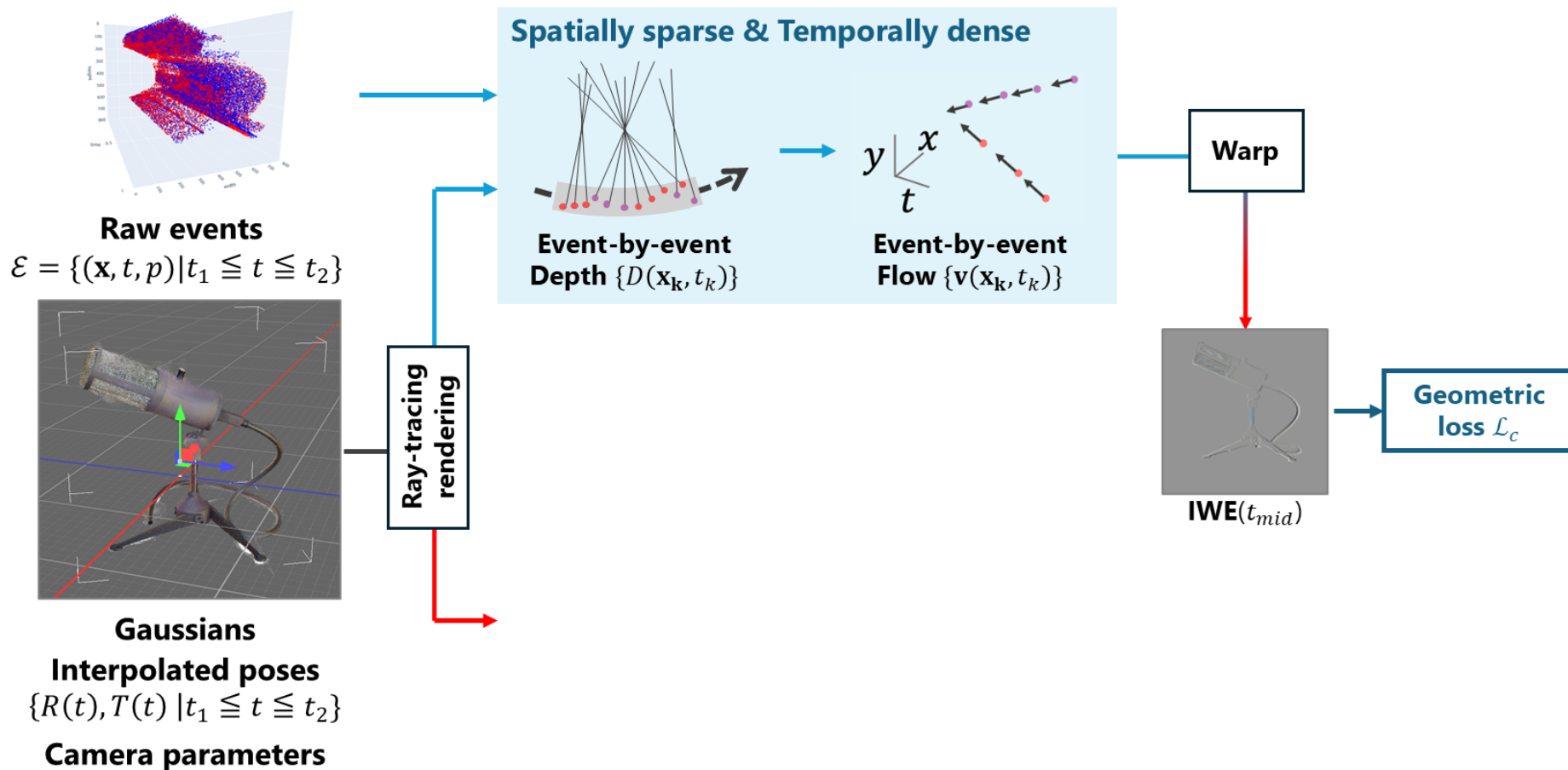


Ground Truth

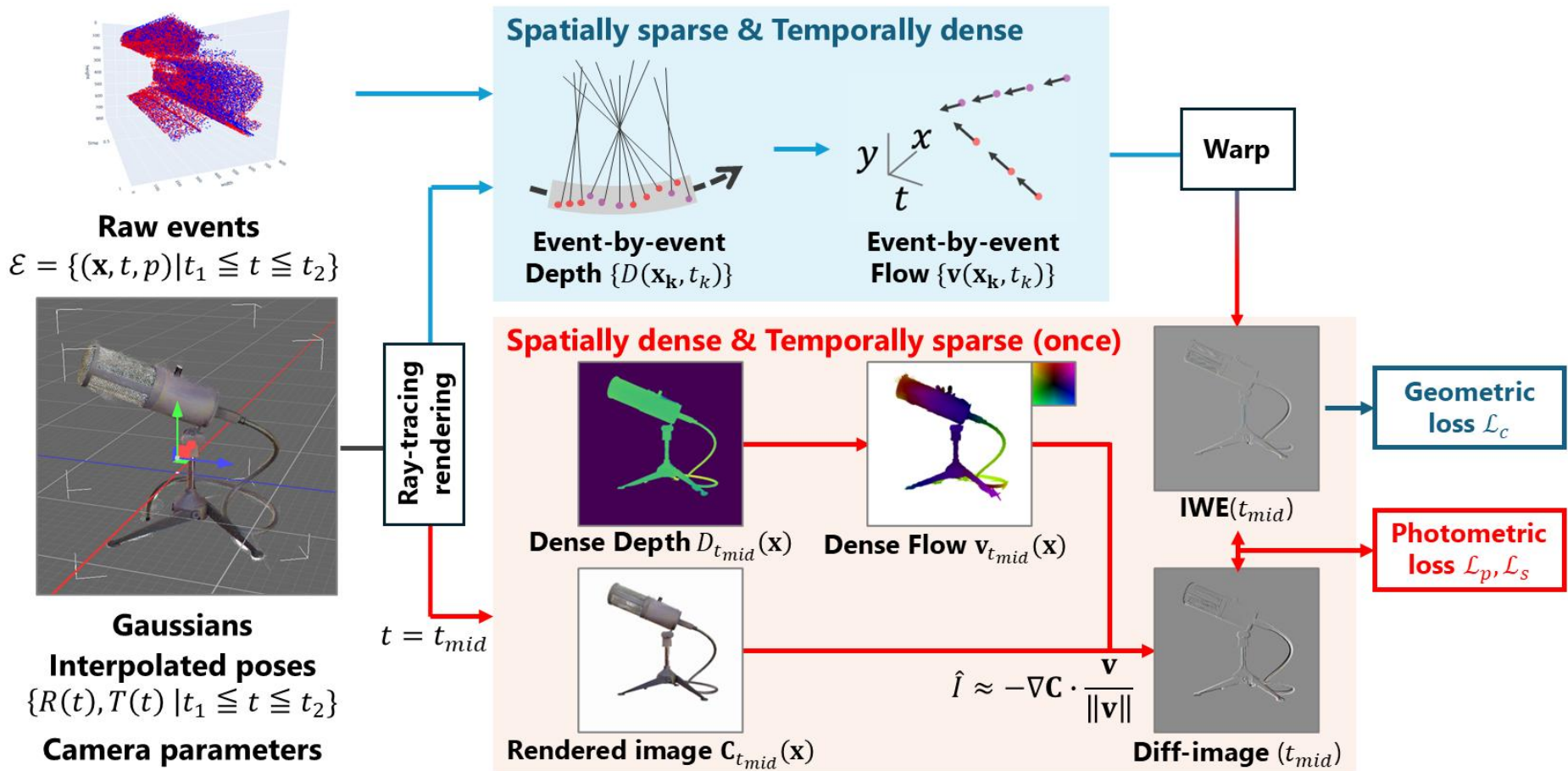
Decompose rendering in two branches



Geometric branch



Photometric branch



Results (VGA & HD data)

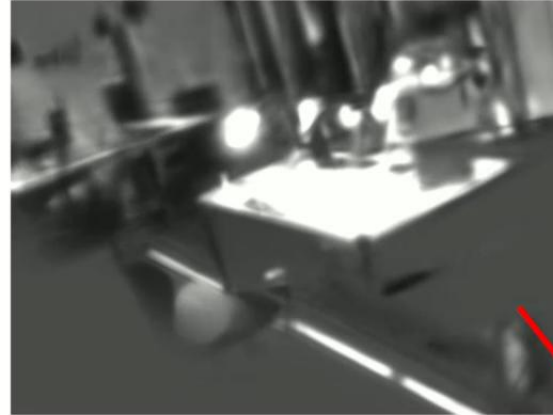
Results

EDS dataset
sequence: 08
640 × 480 px



Noisy

Robust e-NeRF
(event-by-event)



Blurry

IncEventGS
(rendering two frames)



Ours



Ground Truth

Results (synthetic color dataset)

Results

Robust e-NeRF dataset
sequence: chair
800 × 800 px (color event)



Robust e-NeRF



Ours

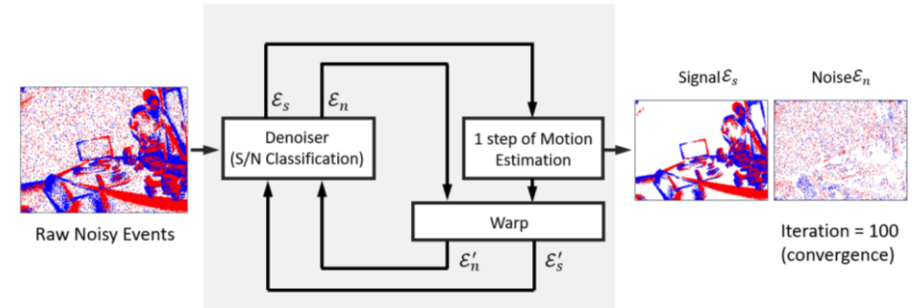
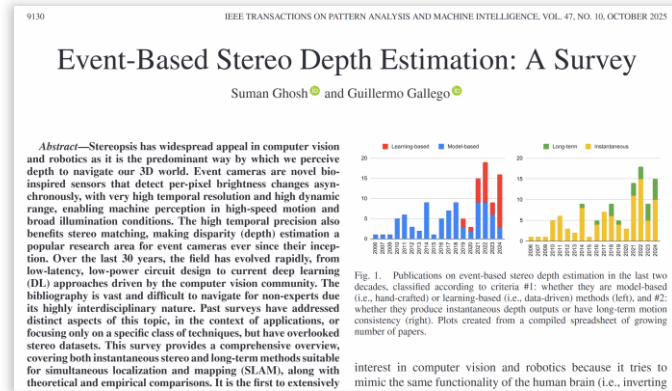


Ground Truth

Insights

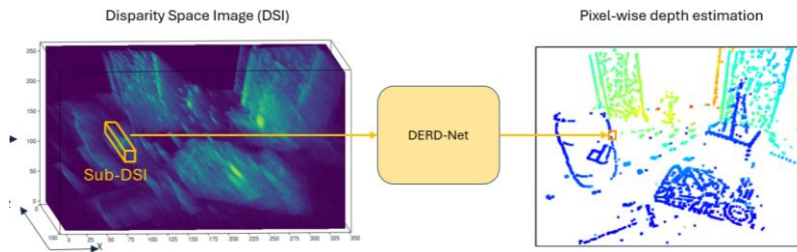
- Event Cameras are a **natural fit** for motion and depth estimation tasks, with **opportunities** across many disciplines.
- **These estimation** tasks (optical flow, SLAM, etc.) allow us to exploit the **high temporal resolution** of the events.
- We seek **unifying theory** or **principles** (CMax and PhE) of event processing to use across problems that:
 - alleviate data annotation (**unsupervised** solutions)
 - can be used with Learning-based methods.
- We seek **joint estimation** of multiple quantities, for robustness and to obtain a comprehensive perception.
- There are **many open problems**
 - **Event data** is completely different from what we are used to.
 - Goal: achieve **fast-and-robust perception**, as promised by neuromorphic systems.

Summary

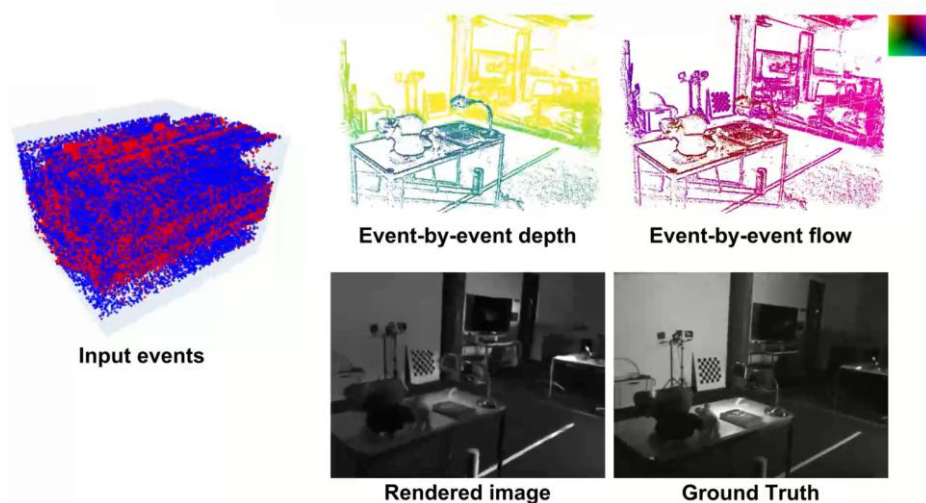


Event-based Stereo Survey, T-PAMI 2025.

ESMD, ICCV 2025.



DERD-Net, NeurIPS 2025.



E2FAI, ICCV 2025 & GPRT, CVPR 2026.

Resources

- **Research at TUB-RIP Lab**

<https://sites.google.com/view/guillermogallego/research/event-based-vision>

- **Course** from TU Berlin: Event-based Robot Vision

<https://tinyurl.com/82wk7wkm>

- **Overview paper** (TPAMI 2022)

<http://rpg.ifi.uzh.ch/docs/EventVisionSurvey.pdf>

- **New Survey paper:**

[*Event-based Stereo Depth Estimation: A Survey*](#) (TPAMI 2025).

- **List** of Event-based Vision Resources

https://github.com/uzh-rpg/event-based_vision_resources

- Review **papers**, **code**, **datasets**, etc.
- **Workshops** (videos, slides)